

Response Letter with Annotations

Wang, L., Yang, A., & Thorson, K. (2021). Serial participants of social media climate discussion as a community of practice: a longitudinal network analysis. *Information, Communication & Society*, 24(7), 941–959. <https://doi.org/10.1080/1369118X.2019.1668457>

Annotations

1. Purposes

- a. Motivation (one-sentence summary): To understand whether climate serial participants (who talk about climate themes) on Twitter form a real community of practice (CoP - serial participants who interact with each other) and how this CoP sustains the movement over time.
- b. Specific Rationale (several bullet points):
 - They collect historical tweets with the keyword “climate” using the data mining service Sifter, identify 12 high-attention “spike” events over 5 years, and then pick the 5 events with the most tweets (Hurricane Sandy, US–China G20 climate announcement, Pope’s encyclical, People’s Climate Movement, Obama’s climate Facebook post).
 - For each of these 5 events they buy the full firehose of “climate” tweets (1,023,863 tweets total) and build mention networks, because a mention reflects prior awareness and is more relational than a retweet.
 - They first find more than 1000 accounts that tweeted in all 5 events (sustained commitment), then restrict to those that interacted (mention) with another serial participant in every event, creating a panel of 290 serial participants over 5 time waves.
 - They run a content analysis of the 290 accounts to code a binary organization/individual account variable and a detailed serial participant classification (mainstream media, online-only media, journalists, advocacy groups, political and governmental actors, celebrities, educators/scientists, etc.), with reported Krippendorff’s alpha values (measure of agreement among human coders).
 - Using the mention networks, they compute standard network centrality measures (degree, closeness, betweenness, eigenvector) for each serial participant and also for everyone in the broader general public network of “climate” tweeters in each event. They build mention networks (@ interactions) where nodes are users and edges are mentions.
 - To understand the network, they calculate centrality measures to capture each actor’s position and importance: degree (number of connections), closeness (how close to all others), betweenness (bridge between

groups), and eigenvector (connection to influential actors). This is done for both the serial participant CoP network and the general public network.

- To study network evolution and CoP structure (RQ2 and RQ3), they apply a stochastic actor-oriented model (SAOM).
- The SAOM (RSiena package on R) model is a longitudinal network model that explains how and why a network changes over time based on the choices of individual actors. It assumes that people in a network actively choose who to connect with (or disconnect from), and these small decisions gradually shape the structure of the network.
- With SAOM, they modeled some effects as social selection mechanisms shaping the mention network between waves 2–5.
- To test whether being central in the CoP increases influence in the broader network (RQ4), they estimate mixed-effects regression models for longitudinal data (LMM, lme4) with eigenvector centrality (how influential you are) and closeness centrality (how easily you connect with others) in the general public network as dependent variables, predicted by serial participants' positions in the serial participant CoP, time, number of followers/friends, and account characteristics (organization/individual, actor type).
 1. LMM is a regression model for longitudinal data that accounts for both fixed effects and individual-level random variation.
 2. Lme4 is the R package
- As a conclusion, serial participants' stable community of practice on Twitter not only develops clear hierarchies (favoring organizational actors) but also provides “relational value” because central positions inside this CoP significantly boost members' elite connections and brokerage roles in the larger public climate discussion network.

c. Research Questions and/or Hypotheses

- Identify IV, DV, mediator, moderator
 1. Dependent variable (DV): Tie formation, the network structure over time among serial participants (who mentions whom in each wave)
- Independent variables (IVs):
 1. Endogenous effects that capture structural patterns in the network, such as reciprocity (mutual ties), transitive triplets (“friend of a friend” in a triangle), and three-cycles (hierarchical patterns).
 2. Exogenous effects that capture serial participants attributes, such as organization vs. individual and actor type, including ego (sender), alter (receiver), and similarity (homophily).
- Moderators: The inclusion of interaction terms (e.g., network position x time) suggests conditional effects over time, which can be interpreted as moderation effects, but the study does not explicitly model or theorize moderation.
- Identify specific relationships among variables

1. More central actors in the CoP are more connected to influential actors in the broader network.
2. More central actors in the CoP have more efficient access to the broader network.
3. The effect of CoP position on influence varies over time (interaction effect).
4. More followers are associated with higher centrality, but with a smaller effect than CoP position.
5. Actor characteristics influence who interacts with whom (social selection).
6. Structural tendencies shape the formation and hierarchy of the network.

2. Method

- a. Research Design: The study combines content analysis with social network analysis and longitudinal modeling (SAOM and LMM) to examine the structure, evolution, and impact of the community of practice over time.
 - Content Analysis:
 1. Data source: Twitter data (tweets containing the keyword “climate”)
 2. Sample: 5 climate-related spike events, 290 serial participants
 3. Manual coding of user: profiles organization vs. individual and actor type (media, journalists, advocacy groups, etc.)
 4. Reliability: Krippendorff’s alpha
 - Social network analysis (SNA):
 1. Examines interaction patterns.
 2. To analyze network evolution over time, the study incorporates longitudinal modeling:
 - a. SAOM using RSiena is used to examine how ties form and evolve, including endogenous and exogenous effects.
 - b. Linear mixed-effects models (LMM) to test the impact of CoP positions on influence in the broader network.
- b. Sample
 - It is a purposive sample of Twitter users.
 - Data come from five climate-related spike events (high-attention periods).
 - From these events, the authors identify 290 serial participants
- c. Measures
 - Variables were measured through manual coding (content analysis), network metrics computed from mention networks (SNA), longitudinal network parameters estimated via RSiena (SAOM), and centrality outcomes modeled using mixed-effects regression (LMM).
- d. Analysis
 - Content analysis + SNA + SAOM + LLM

3. Results and Conclusions

a. Answers to RQs and Hypotheses

- RQ1: Who are the participants involved in the climate change serial participant network over time, and do their interactions suggest the development of a CoP?
 1. The study finds that serial participants, including both individuals and organizations, form a community of practice (CoP) through sustained interaction over time. Although the network begins as loosely connected, repeated engagement and mentions among participants lead to a more stable and cohesive structure. A core group of highly active users emerges, indicating the development of a persistent community centered around climate discussions.
- RQ2: Do network structural features (tendencies toward reciprocity, transitivity, and three-cycles) significantly influence the structure of the serial participants' community over time?
 1. The results show that endogenous network structures significantly shape the evolution of the CoP. Positive reciprocity indicates that participants tend to return interactions, while positive transitive triplets suggest the formation of clustered relationships. At the same time, negative three-cycles reveal hierarchical patterns within the network, and negative outdegree effects indicate that users are selective in forming ties. Together, these findings demonstrate that the network is structured and self-organizing rather than random.
- RQ3: Do serial participants' actor attributes (organizational vs. individual accounts, type of participants) significantly influence the structure of the serial participants' CoP over time?
 1. The study finds that actor attributes influence tie formation through exogenous effects. Organizational accounts are more likely to be selected as interaction partners and tend to connect with other organizations, while individuals are less likely to receive reciprocated ties. Although actor type does not strongly predict popularity, similarity between actors affects interaction patterns. These results suggest the presence of social selection mechanisms and status differences within the network.
- RQ4: How do serial participants' network positions in their CoP affect their network positions in the larger Twitter conversation about climate?
 1. The findings show that participation in the CoP provides clear advantages in the broader network. Serial participants who occupy central positions within the CoP are more likely to achieve higher eigenvector centrality and closeness centrality in the general public network. This means they are better connected to influential actors and have more efficient access to the network. While the number of followers also contributes to centrality, its effect is smaller than that of CoP position, indicating that network position within the community plays a key role in shaping influence over time.

b. Include Key Figures/Tables

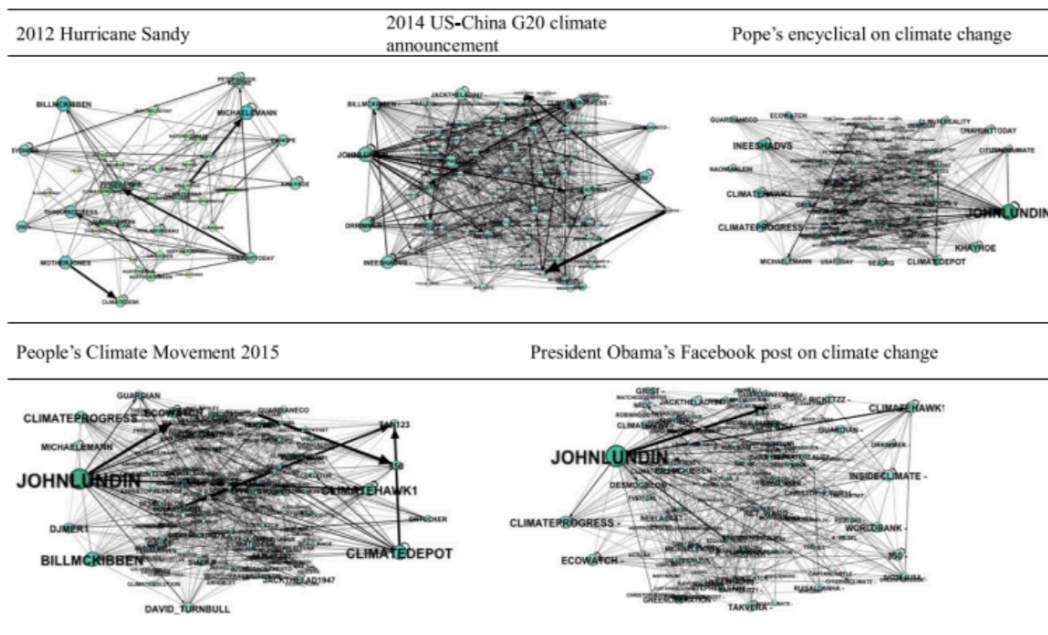


Figure 1. Serial participant mention networks in five activism events.

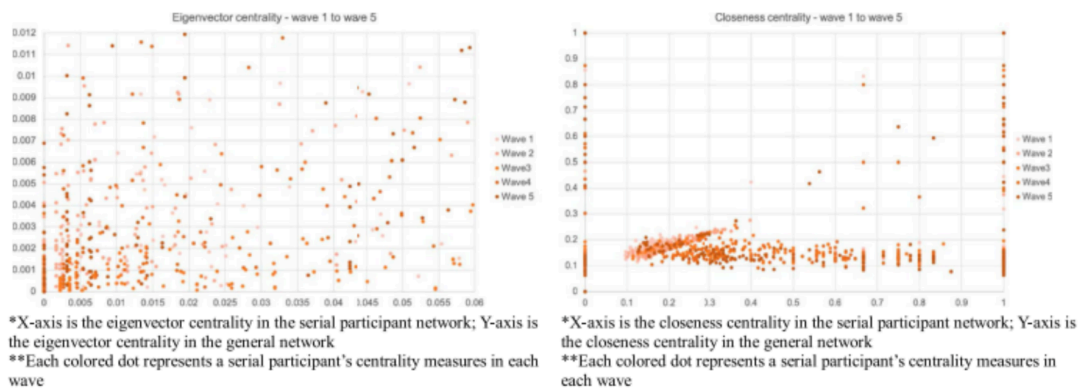


Figure 2. The changing dynamics of serial activists' eigenvector centrality and closeness centrality.

Table 2. Top 20 (weighted degree) serial users categorization at five spike events.

Spike events	Organization vs. individual accounts	Top type of serial user accounts
2012 Hurricane Sandy	Individuals (7); Organizations (13)	Online only media (5) Advocacy groups (3)
2014 US China G20 climate announcement	Individuals (13); Organizations (7)	Individual activists (7)
Pope's encyclical on climate change	Individuals (19); Organizations (1)	Individual activists (12)
People's Climate Movement 2015	Individuals (14); Organizations (6)	Individual activists (5) Advocacy groups (3)
President Obama's Facebook post on climate change	Individuals (9); Organizations (11)	Advocacy groups (5) Individual activists (4)

Table 3. Results of the SIENA multivariate network analysis: estimates of between-network effects for friendship and antipathy controlled for structural network effects, and individual and behavioral dispositions.

	Estimate	Standard error	Convergence t-Ratio	<i>p</i> *
<i>Rate constant</i>				
Serial activist network rate (period 1)	19.48	(1.77)	−.11	**
Serial Activist network rate (period 2)	10.69	(.86)	.01	**
Serial Activist network rate (period 3)	8.30	(.77)	−.05	**
<i>Structural network effects</i>				
Outdegree (density)	−2.95	(.03)	−.10	**
Reciprocity	1.27	(.11)	−.01	**
Transitive triplets	.68	(.03)	−.08	**
Three-cycles	−2.23	(.62)	−.08	**
<i>Serial activist characteristics</i>				
Organization/individual alter	.37	(.03)	.02	**
Organization/individual ego	−.33	(.05)	.01	**
Organization/individual similarity	.04	(.05)	−.08	n.s.
Actor type alter	.05	(.03)	.06	n.s.
Actor type ego	.07	(.04)	.02	n.s.
Actor type similarity	−.3479	(.06)	−.09	**

*The significance of these evaluation function weights was assessed by dividing the estimates by their standard errors. Any absolute values greater than 2 are significant at the .05 level (Snijders et al., 2013).

p* < .01, *p* < .05; ****p* < .001.

c. Authors' Conclusions

- Serial participants in climate discussions on Twitter form a community of practice (CoP) through sustained interaction over time. This community evolves from a loosely connected group into a more stable and structured network, driven by social selection mechanisms.
- Network dynamics are shaped by both endogenous structural effects and exogenous actor attributes, leading to the emergence of hierarchies within the network. Organizational actors tend to occupy more central and preferred positions, reinforcing status differences among participants.
- Participation in the CoP provides functional value. Actors who are central within the CoP gain greater visibility, influence, and access to elite connections in the broader public network. Thus, even though the

CoP is relatively small, it plays a critical role in shaping influence in large-scale public discourse.

- Sustained interaction in social media environments can lead to the formation of structured and influential communities, highlighting the importance of long-term engagement in digital activism and issue communication.

4. Discussion

a. Implications

- The study shows that sustained interaction on social media can lead to the formation of a community of practice (CoP) that is structured, hierarchical, and influential. This challenges the idea that online participation is mostly superficial or short-lived.
- It highlights the role of social selection mechanisms, demonstrating that both network structure and actor attributes shape interaction patterns. In particular, the prominence of organizational actors suggests that traditional forms of authority and legitimacy continue to matter in social media environments.
- Being central in a CoP increases visibility, access to elites, and influence in the broader public network. This suggests that small, highly engaged communities can have a disproportionate impact on public discourse.

b. Limitations

- I don't think the sample is representative, as it focuses on a purposive group of highly active serial participants rather than the general population of Twitter users.
- The analysis relies on Twitter mention networks, which capture only one type of interaction and may not reflect all forms of engagement (e.g., retweets, likes, offline actions).
- The study focuses on a single issue (climate change) and a specific platform, which may limit the generalizability of the findings to other contexts or topics.
- Finally, although the study uses longitudinal modeling, it remains observational, meaning causal relationships cannot be definitively established.

c. Future Directions

- Examine how opposing groups interact with or influence communities of practice in online discussions.
- Explore whether similar patterns emerge in other issues, platforms, or cultural contexts, testing the generalizability of the findings.
- Investigate other forms of interaction (e.g., retweets, replies) or incorporate.
- Analyse power dynamics and inequalities within CoPs, including how certain actors gain or maintain central positions over time.

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Response Letter (~600 words)

- Your assessment of the study

Overall, this study offers a strong contribution to the understanding of online communities and digital activism, particularly through its application of the community of practice (CoP) framework to social media environments. In my perspective, one of its main strengths lies in the integration of multiple methods, content analysis, social network analysis (SNA), and longitudinal modeling (SAOM and LMM), which allows the authors to examine not only the structure of the network but also its evolution and broader impact. This multi-method approach strengthens the validity of the findings and provides a more comprehensive understanding of how sustained participation shapes influence in online discourse.

Another key strength is the longitudinal design. By analyzing multiple time waves, the study captures dynamic processes such as tie formation, stability, and change over time. This is particularly important in the context of digital communication, where interactions are fluid and rapidly evolving.

The study also makes an important theoretical contribution by demonstrating that online participation can lead to the development of structured and hierarchical communities, challenging the common assumption that social media interactions are fragmented or superficial. The finding that centrality within a small CoP predicts influence in the broader network is particularly compelling, as it highlights how relatively small groups can shape large-scale public discourse.

- Weaknesses

Despite its strengths, the study has several limitations. First, the sample is not representative, as it focuses on a specific group of highly active “serial participants.” While this is appropriate for the research question, it limits the generalizability of the findings to the broader population of social media users. The behaviors and dynamics observed among highly engaged users may differ significantly from those of more typical or casual participants.

Second, the study relies exclusively on mention networks, which capture only one form of interaction. Other important forms of engagement, such as retweets, likes, or replies, are not included in the analysis. This may result in a partial view of the communication network, potentially overlooking different types of influence or interaction dynamics.

Third, although the study uses sophisticated longitudinal models, it remains observational. As a result, it cannot establish causal relationships between variables. For example, while centrality in the CoP is associated with greater influence in the broader network, it is not possible to definitively conclude that one causes the other.

Additionally, while the study incorporates actor attributes, it does not deeply explore contextual or content-based factors, such as message framing or ideological differences, which could also influence interaction patterns.

- Other notes

One interesting extension of this study would be to apply a similar network approach to the spread of misinformation, particularly among communities such as anti-vaccine groups. The concept of a community of practice (CoP) could be useful to understand how highly active users who consistently share misinformation form stable and influential networks over time.